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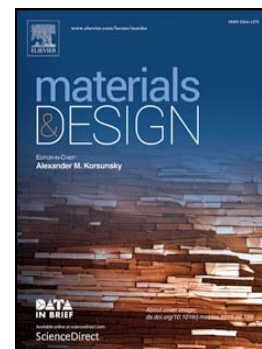
Biodegradable hybrid nanocomposites of starch/lysine and ZnO nanoparticles with shape memory properties

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BIODEGRADABLE HYBRID NANOCOMPOSITES OF STARCH/LYSINE AND ZNO NANOPARTICLES WITH SHAPE MEMORY PROPERTIES

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ABSTRACT

In food packing applications; the research works have been focused on the development of biodegradable packing materials by using biopolymers such as starch, carrageenan, agar, and gelatine to replace with petroleum-based non-biodegradable polymers. In this present work we studied novel synthesis of biodegradable hybrid nanocomposites using starch, amino acid (Lysine), polypropylene glycol (PPG) and ZnO nanoparticles (NPs) and their shape memory properties. A series of hybrid nanocomposites were prepared by solution casting method, with various ZnO nanoparticles content such as 0 wt%, 1 wt%, 3 wt% and 5 wt% by keeping peptide content constant and varying the ratio between starch and ZnO nanoparticles content. The thermogravimetric analysis (TGA) of the hybrid nanocomposites suggested increased thermal stability with increasing ZnO nanoparticles content. The solubility of the hybrid nanocomposites in H₂O is decreased with increasing ZnO nanoparticles content. The moisture content in the hybrid nanocomposites is decreased with increasing ZnO nanoparticles content. The mechanical properties of the materials were increased with increasing ZnO nanoparticle content. Furthermore, the hybrid nanocomposites showed a shape memory properties by placing the sample at room temperature and then at 55°C. The morphology of the hybrid nanocomposites was revealed using scanning electron microscopy (SEM). We believe that these biodegradable hybrid nanocomposite films can be useful in food packing applications.

Key words: Rice Starch; Lysine; Polypropylene glycol; ZnO Nanoparticles; hybrid nanocomposites; shape memory effects.

1. Introduction

The food packaging materials are always required with proper mechanical strength, excellent barrier properties against water, water vapor, and other gases, thermal stability, recyclability, and biodegradability as well as functional properties such as antioxidant activities and antimicrobial are highly desirable for food safety and extending the shelf-life of packaged foods [1]. The development of new biodegradable packaging materials from environmental sources and renewable raw materials is an inevitable and interesting research area. Currently the materials that have been used in packaging industries are mainly prepared by using petroleum based plastic materials, since they are relatively cheap and convenient to use with excellent process ability and durability [2, 3].

However, this petroleum based plastic materials has a drawback that is non-biodegradable in nature, the soaring price of oil, and concerns on the exhaustion of natural resources. Indeed, from the past few years, researches have been focused on the development of biodegradable packaging materials using polymers to replace petroleum-based plastic packaging materials. Until now, a variety of biopolymers such as starch, carrageenan, agar, and gelatine have been used for making biodegradable plastics [4]. However, due to their poor barrier, mechanical and processing properties with high production costs compared with the petroleum based plastic materials have been the main limitation in using the biopolymer films for food packaging application [5]. Generally, natural polymers and biopolymers are used to blend with other synthetic polymers or nanomaterials in order to improve their properties and applications.

Biodegradable green polymer films are generally designed using biomass such as proteins, polysaccharides, lipids, and their derivatives [6, 7, 8, 9, 10]. Among these biopolymers,

starch is an ideal polysaccharide, inexpensive, renewable, non-toxic, environment-friendly material and biodegradable natural biopolymer. It contains two microstructures, e.g. linear (amylose) and branched (amylopectin). Amylose has a linear structure of α -1,4 linked glucose units with a molecular weight of 105–106 and amylopectin is a highly branched structure of short α -1,4 chains linked by α -1,6 bonds with a molecular weight of 107–109, both of which are polymers of α -D-glucose units, a highly branched polymer being used in the production of biodegradable packaging materials due to its low cost, biodegradability, biocompatibility, and ease of chemical derivatization [11,12,13,14,15]. Besides, the plasticizers such as glycerol, propylene glycol and sorbitol used to enhance the flexibility of the biopolymer films are also stated to impair the mechanical and barrier properties [16,17]. However, wide applications of starch-based films have been limited due to their poor mechanical properties, moisture sensitivity and brittleness [18].

Amino acids are the monomeric units of proteins with a relatively low molecular weight and a range of different properties depending on variations in their side chains (which can be acidic, basic, hydrophobic, etc.). It has been reported that the introduction of amino acid moieties to the backbone of chitosan shown to improve the physicochemical properties (solubility and mucoadhesiveness) of chitosan and gives rise to some interesting synergistic characteristics for use in drug delivery and tissue engineering, as well as other potentially useful biomedical properties such as anticoagulant, antimicrobial and anti-cholesterol activities [19]. Lysine is an amino acid and widely used in biomaterials and it have been incorporated into peptides are termed "residues" due to the release of either a hydrogen ion from the amine end or a hydroxyl ion from the carboxyl end, or both, as a water molecule is released during the formation of each amide bond. Poly(L-lysine) (PLL) is widely used for

the constructions of biomaterials for several applications such as tissue engineering and drug delivery [20, 21].

The incorporation of inorganic nanoparticles into the polymeric matrix to enhance the properties such as thermal and mechanical properties of polymeric materials is well known. Several nanoparticles, including nanoclays, nano metal or metal oxide have been used to improve the polymer properties [22]. Recently, utilization of nano metal or metal oxides, such as silver, ZnO, CuO, and TiO₂ as reinforcing filler in polymers has emerged due to their high thermal stability and certain functions like strong antimicrobial activity [23, 24]. Among the several metal oxide, ZnO nanoparticles (ZnO NPs) have large specific surface area and high surface energy and have been extensively used in nanocomposites due to the excellent ability at nano-scale dispersion and interfacial interactions in a protein matrix. In biopolymer films (such as starch based films) the ZnO can be incorporated to improve mechanical and water vapour barrier properties. In addition, they are believed to be nontoxic and listed as generally recognized as safe [25].

Therefore, in the present study, the special attention has given for the selection of materials such as renewable, biocompatible and microbial activities in order to create novel materials. Then proceed in the development of novel biodegradable hybrid nanocomposites by using starch, lysine and ZnO NPs in order to improve the thermal, mechanical and barrier properties of the materials and to use them in food packing applications as well as in biomedical applications.

2 Experimental and Methods

2.1 Materials

All raw materials used in this study are available as commercial products. Rice starch was obtained from S.D Fine Chemicals Limited, Mumbai, India. Lysine was purchased from Labcon chemicals limited South Africa. EP grade Propylene Glycol and ZnO nanoparticles was purchased from Sigma Aldrich Chemicals, South Africa.

2.2 Preparation of hybrid nanocomposites

The hybrid nanocomposite films were produced by using solution casting method described below. Film-forming solutions were made by dispersing RS (control), Lysine and ZnO nanoparticles containing different contents (0 wt%, 1 wt%, 3 wt%, 5 wt%) based on the total mass of dry solid (5 g) and propylene glycol (20 % wt.) in 100 ml of distilled water as shown in Table 1. To achieve hybrid nanocomposites, the mixture was stirred constantly at 90 °C for 60 min at 1500 rpm using a magnetic stirrer on a hot plate. Film thickness was controlled by measuring the volume of film-forming solution. The film forming solutions (100 ml) was cast on Teflon plates and cured at $(55 \pm 1)^\circ\text{C}$ for 4-6 h with circulating air. The obtained films were removed from the plates and kept at $25 \pm 1^\circ\text{C}$ in a desiccator containing $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ ($50 \pm 2\%$ RH) for 72 h before characterization. The processing path of the film casting is shown in Scheme. 1.

2.3 Morphology

The hybrid nanocomposite films were dried in vacuum and stored in a desiccator before the SEM micrographs recorded. The dry samples were gold-coated by an electro deposition

technique to impart electrical conduction. The morphology of the films was studied using JEOL JEM-7500F (Tokyo, Japan) operated at an accelerating voltage of 2 kV.

2.4 Thermal Properties

Thermogravimetric analyses (TGA) of the prepared hybrid nanocomposites films were carried out using SDT Q 600TA Instruments. The specimens were scanned under N₂ atmosphere; from room temperature to 600 °C at a heating rate of 10 °C/min.

2.5 Mechanical Properties

Tensile tests were performed as per ASTM D 882-91 standard. Specimens were cut into rectangular shape with dimensions 100 x 20 x 0.2 mm. Tensile properties such as the maximum stress and % elongation at break were determined using an MTS-Universal Testing Machine at a crosshead speed of 50 mm/min maintaining a gauge length of 50 mm. In each case five specimens were tested and the average values are reported.

2.6 Moisture Content and Water solubility

The moisture content in the film was determined by measuring the weight, upon drying films in an oven at 110 °C until no further reduction of weight was observed. Three replicates of film specimens were used to measure the moisture content. The film solubility method was adapted from the literature [26]. The initial dry matter of each film was obtained after drying film specimens at 60 °C for 24 h. These dried films (about 0.3 g) were weighed (initial dry weight) and immersed in beakers containing 50 mL distilled water at 23°C that were then sealed and periodically agitated for 24 h. Tests were performed in triplicate and the solubility was calculated using Equation. 1.

Initial weight - Final weight

Solubility (%) = ----- x 100.....Eq. 1.

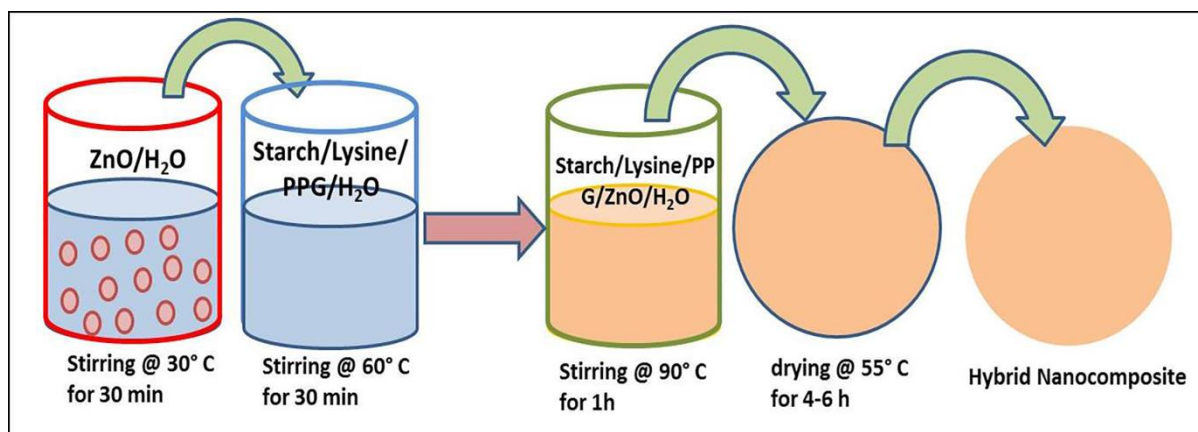
Initial weight

2.7 X-ray Diffraction (XRD)

XRD patterns of the samples were recorded on a X-ray diffractometer (PANalytical-Empyrean theta-theta) equipped with Co (Anode material) at 40 kV and 40 mA. Data was collected from 3 degree (°) to 90 degree (°) Bragg angles (2θ) at a scan rate of 3 degree (°)/min.

3. Results and Discussions

In order to develop a novel hybrid materials for food packing applications special attention has been given for the selection of materials such as renewable, biocompatible and microbial activities. Then proceed in the development of novel biodegradable hybrid nanocomposites by using starch, lysine and ZnO NPs in order to improve the thermal, mechanical and barrier properties of the materials. A series of hybrid nanocomposites were prepared by solution casting method as shown in Scheme. 1, with various ZnO NPs content such as 0 wt%, 1 wt%, 3 wt% and 5 wt% by keeping peptide content constant and varying the ratio between starch and ZnO NPs content. The chemical compositions used in this study are given in the Table. 1.



Scheme. 1. Schematic illustration of the Preparation of hybrid nanocomposite films.

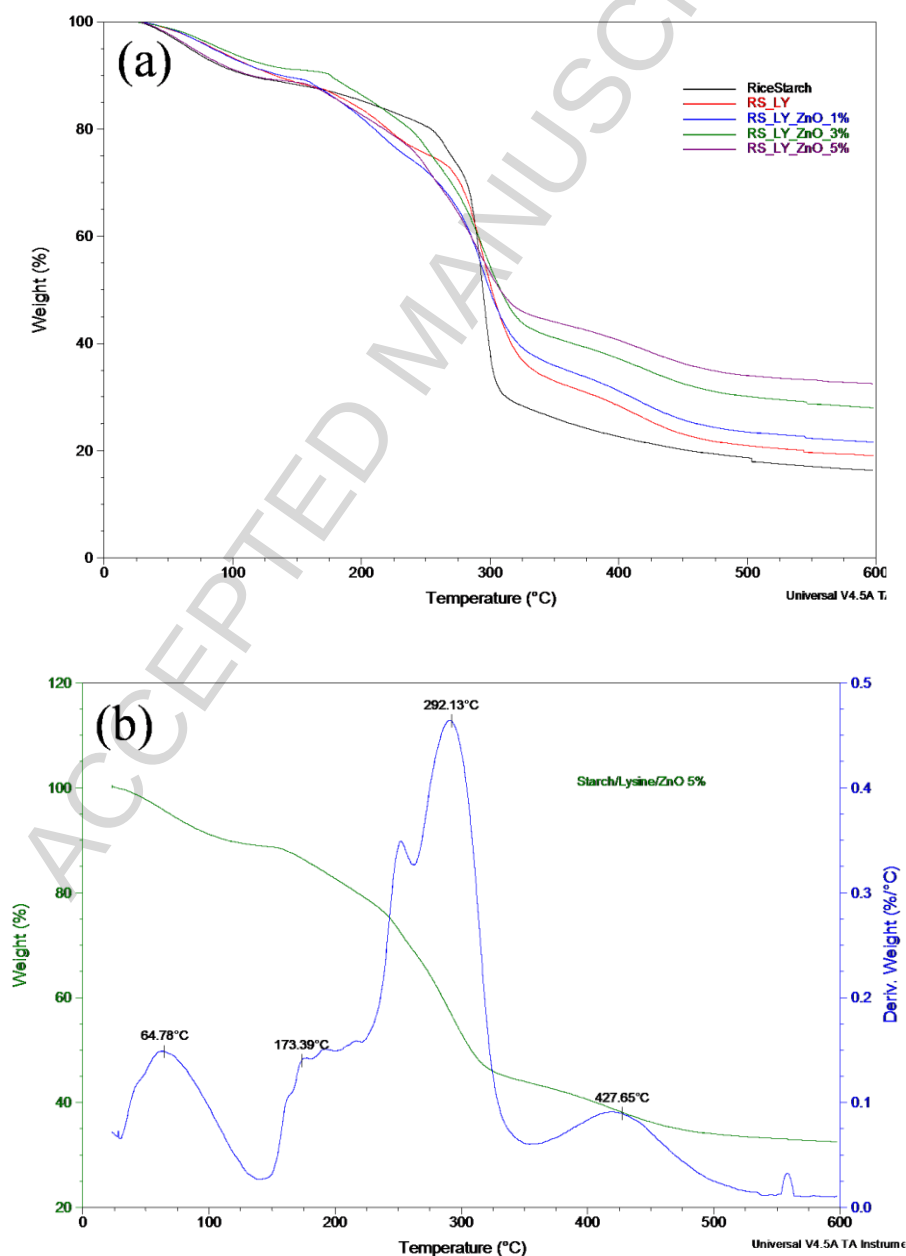
Table 1. Chemical compositions used for the preparation of hybrid nanocomposite films.

	Starch (g)	Lysine (g)	ZnO NPs (g)	Polypropylene glycol (g)	H ₂ O (mL)
Starch	5.000	0.000	0.000	1.500	100
Starch/Lysine	4.000	1.000	0.000	1.500	100
Starch/Lysine/ZnO 1 wt%	3.950	1.000	0.050	1.500	100
Starch/Lysine/ZnO 3 wt%	3.850	1.000	0.150	1.500	100
Starch/Lysine/ZnO 5 wt%	3.750	1.000	0.250	1.500	100

3.1 Thermogravimetric analysis

Thermogravimetric analysis of the hybrid nanocomposite films showed four steps of weight loss, one is at about 50-150°C due to the physisorbed water, which is of very less percentage. The weight loss of 150-250°C, 250-350°C and 350-500°C is ascribed to the removal of organic functional groups of starch and lysine. In all the films maximum weight loss is observed nearly at 295°C. At 600°C the ZnO NPs filled nanocomposites films behavior is similar to starch. The increased residue from 16 % to 32 % is observed at 600°C with

increasing ZnO NPs content as shown in Fig. 1a. The thermal stability of the various nanocomposite films is shown in Table 2. The second derivative of the nanocomposite films is suggested that the clear four steps of weight loss as shown in Fig. 1b and Fig. 1c. In starch film, three steps of weight loss are observed as shown in Fig. 1d and confirming that the absence of lysine.



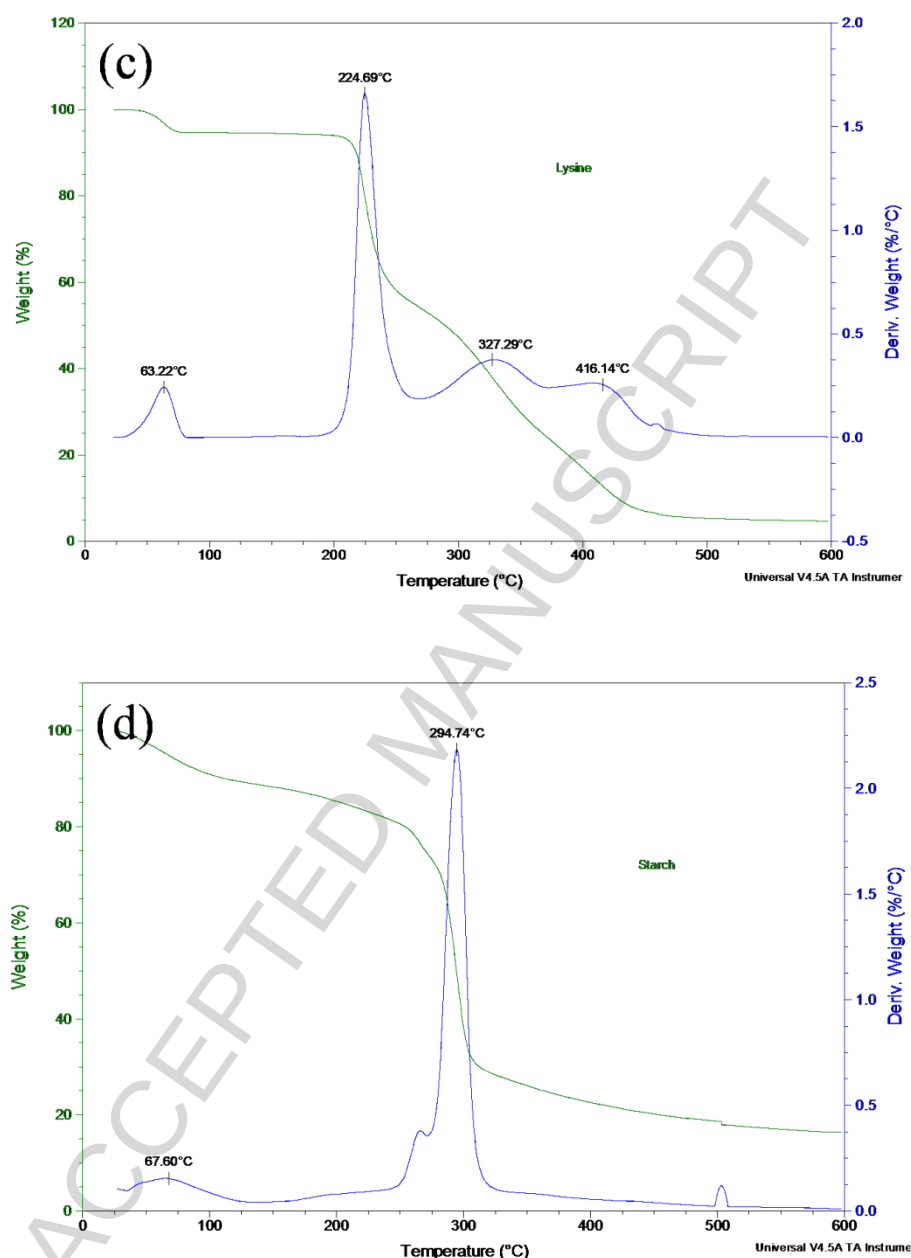


Fig. 1. Thermogravimetric curves of the samples; (a). Overlay of the nanocomposites films, (b). Second derivative of the starch/lysine/ZnO 5%, (c). Second derivative of the lysine and (d). Second derivative of the starch.

Table 2. Thermal stability of the hybrid nanocomposite films.

Sample Configuration	Maximum degradation temperature °C (T_{max})	Residue at 600°C (Wt. %)
Rice Starch	294.74	16.39
RS/Lysine	293.81	19.10
RS/Lysine/ZnO 1 wt%	294.04	21.66
RS/Lysine/ZnO 3 wt%	295.26	28.04
RS/Lysine/ZnO 5 wt%	292.13	32.55

3.2 Mechanical Properties

The mechanical properties of the nanocomposites films were examined using DMA instrument. The results suggested that the strength (modulus) of the materials is increased with increasing ZnO NPs content as shown in Fig. 2a. Stach/Lysine film (ZnO NPs 0 wt%) showed 34.32 MPa, Stach/Lysine film (ZnO NPs 1 wt%) 50.88 MPa, Stach/Lysine film (ZnO NPs 3 wt%) 78.69 Mpa, Stach/Lysine film (ZnO NPs 5 wt%) 96.55 MPa. Furthermore the elongation at break of the nanocomposite films were decreased with increasing ZnO NPs content as shown in Fig. 2b. Starch/Lysine film (ZnO NPs 0 wt%) showed the 42%, Stach/Lysine film (ZnO NPs 1 wt%) 31%, Stach/Lysine film (ZnO NPs 3 wt%) 20%, Stach/Lysine film (ZnO NPs 5 wt%) 13%.

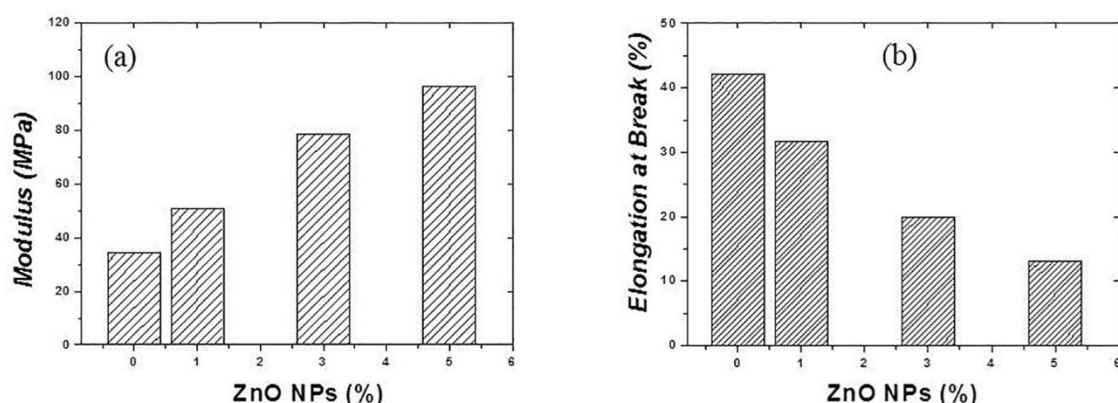


Fig. 2. Mechanical properties of the hybrid nanocomposites films. (a) Modulus of the nanocomposite films. (b) Elongation at break of the nanocomposite films.

3.3 Biodegradability and moisture content

The biodegradability of the nanocomposite films were studied by keeping the sample in water (100% RH) environment as a function of time as shown in Fig.3. In the case of starch/lysine film the solubility is more comparable to ZnO NPs incorporated films and it is reduced with increasing ZnO NPs content. The solubility of the films; 100 % for the sample starch/lysine (ZnO NPs 0%), 43% for the sample ZnO 1 wt%, 38% for the sample ZnO 3 wt% and 37% for the sample ZnO NPs 5 wt%.

The moisture content of the nanocomposite films is calculated by drying the samples in oven at the 110°C and monitored reduction in the weight loss as a function of time as shown in Fig. 3. The weight loss is reduced from 21 to 15% with increasing ZnO NPs content as shown in Fig. 3. The calculated values are 21% for the sample starch/lysine (ZnO NPs 0 wt%), 18% for ZnO 1 wt%, 16% for ZnO 3 wt% and 15% for the sample ZnO 5 wt%.

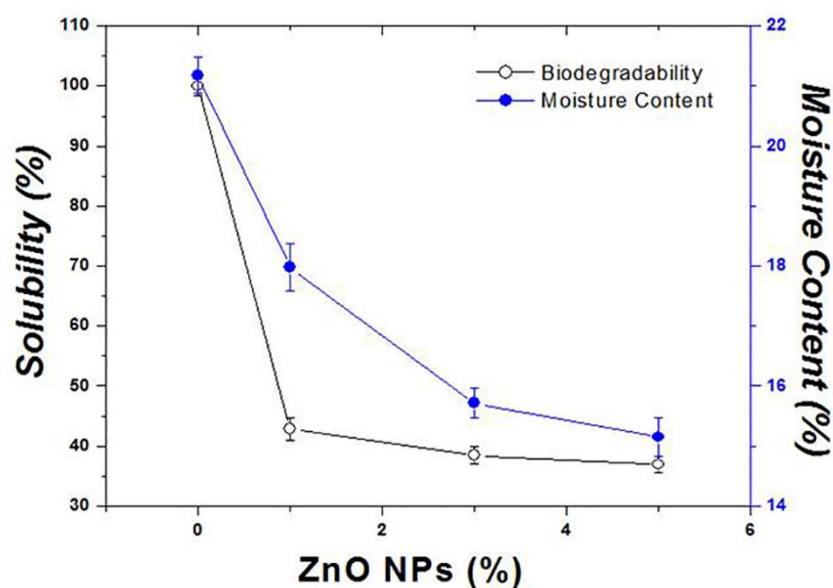


Fig. 3. Biodegradability and moisture content of the hybrid nanocomposite films.

3.4 XRD analysis

Generally, ZnO NPs shows XRD peaks at 2θ values of 37.1, 40.3, 42.5, 55.9, 66.9, 74.6, 79.0, 81.11, and 82.53 due to various crystalline forms of ZnO NPs. From the XRD pattern Fig.4, shows that of Starch/Lysine/ZnO NPs hybrid nanocomposites, it is observed that all the peaks of ZnO NPs (2θ values of 37.1, 40.3, 42.5, 55.9, 66.9, 74.6, 79.0, 81.11, and 82.53) appeared along with the peaks of starch and lysine (2θ values of 16.4, 19.68, 22.66, 25.66, 51.29, 52.26 and 76.22). Interestingly the peak at 2θ values of 14.63, which is observed in Starch film, and the peak at 2θ values of 5.8 and 11.8 which is observed in lysine vanished in hybrid nanocomposite films. This may be due to intercalation and uniform distribution of ZnO NPs within the starch matrix indicating the formation of composites.

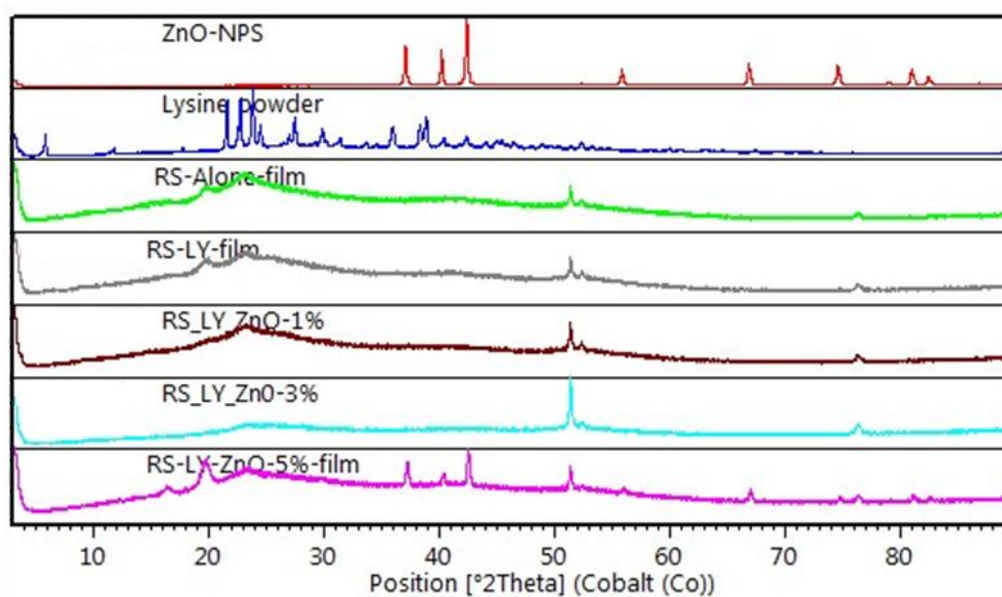


Fig. 4. XRD plots of starch, starch/lysine and the hybrid nanocomposite (starch/lysine/ZnO NPs) films.

3.5 Scanning Electron Microscopy (SEM)

Fig. 5a, 5b, and 5c-5f shows SEM images of starch, starch/lysine and hybrid nanocomposite (starch/lysine/ZnO) films. All of the films showed visibly smooth and compact surface structures similar to the microstructure of cassava starch-gum bio-based films [27]. The SEM image of RS film (Fig.5a) showed a relatively smooth and continuous cross section without pores and cracks, which confirmed a dense and homogeneous structure. The micrographs of the starch/lysine film exhibited a patchy and rough cross section (Fig. 5b), which may occur because of the flexibility of lysine films. In the case of hybrid nanocomposites (starch/lysine/ZnO), noted to improve the internal structure of starch/lysine films, in which RS-lysine hybrid nanocomposite films showed less roughness and more compact and dense appearance (Fig. 5c-5f). SEM micrographics of starch and lysine films showed compact structures. Hybrid nanocomposites (starch/lysine/ZnO) films presented a more heterogeneous surface which seems to have a good compatibility among the polymers. Possibly, the addition of lysine/ZnO NPs to starch matrix changes the microstructural arrangement of lysine chains that are regrouped in the helycoidal way during films drying.

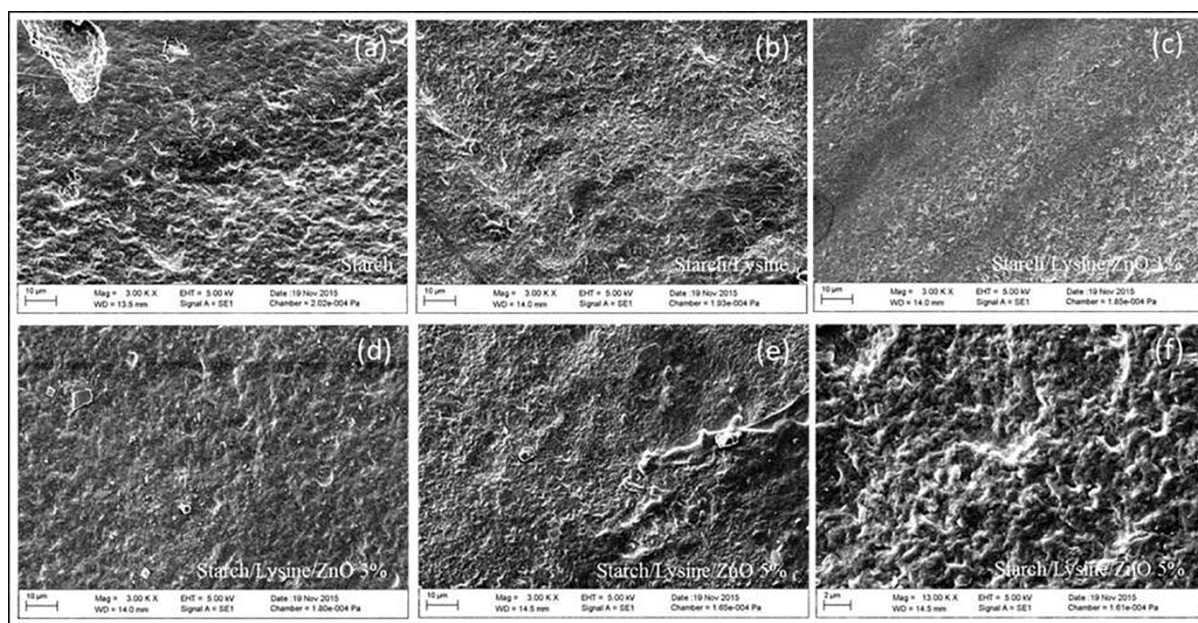


Fig. 5. SEM images of (a).Starch, (b). Starch/Lysine, (c). Starch/Lysine/ZnO 1 wt%, (d). Starch/Lysine/ZnO 3 wt%, (e) and (f). Starch/Lysine/ZnO 5 wt% films.

3.6 Shape memory properties

The hybrid nanocomposite film (ZnO NPs 5 wt%) showed a humidity controlled shape memory behavior as shown Fig.6. Initially at 55°C, over the period of 30 min film molded like a tube. At room temperature in a 100 % humidity environment over the period of 24h film became nearly flat. Finally, at 55°C, over the period of 30 min film reverted back to original shape (like a tube). Our material is behaving as similar as reported recently [28]. Similar behavior is also observed with the hybrid nanocomposites films (ZnO NPs 1 wt%, ZnO NPs 3 wt%).

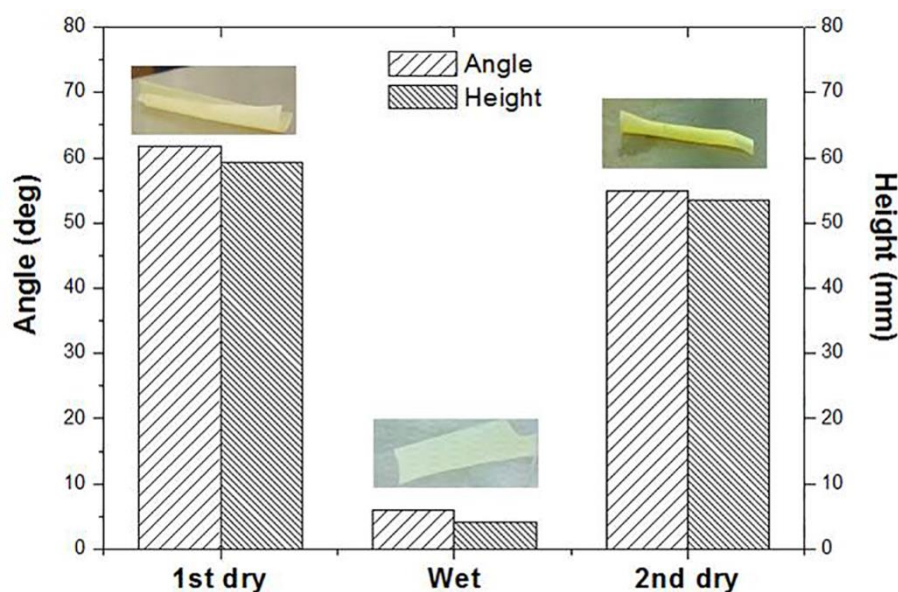


Fig.6. Shape–memory effect of the hybrid nanocomposite films.

4. Conclusion

A series of the biodegradable hybrid nanocomposites were prepared by solution casting method, with various ZnO nanoparticles content such as 0 wt%, 1 wt%, 3 wt% and 5 wt% by keeping lysine content constant and varying the ratio between starch and ZnO nanoparticles content. The thermogravimetric analysis (TGA) of the hybrid nanocomposites suggested that increased thermal stability with increasing ZnO nanoparticles content. The mechanical strength of the materials increased with increasing ZnO NPs content. The solubility and moisture content of the hybrid nanocomposites is decreased with increasing ZnO nanoparticles content. The hybrid nanocomposites showed shape memory properties by placing sample at 55 °C and then at room temperature. We believe that these biodegradable hybrid nanocomposite films can be useful in food packing applications to improve the self-life of the food products.

Acknowledgements

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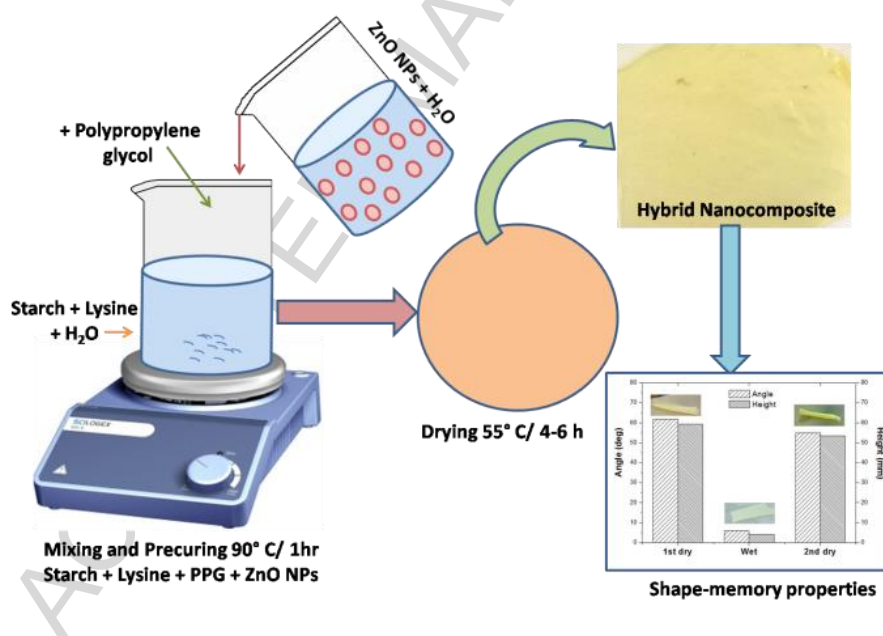
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Highlights

- Novel biodegradable hybrid nanocomposites using starch/lysine/ZnO Nanoparticles.
- Thermal stability of hybrid nanocomposites increased with increasing ZnO NPs content
- The solubility and moisture content of the hybrid nanocomposites is decreased with increasing ZnO NPs content.
- The mechanical properties of the materials were increased with increasing ZnO NPs content.
- Shape memory effect of the nanocomposites.